

ME40064: System Modelling & Simulation

Lecture 1

Dr Andrew Cookson
University of Bath, 2025-26

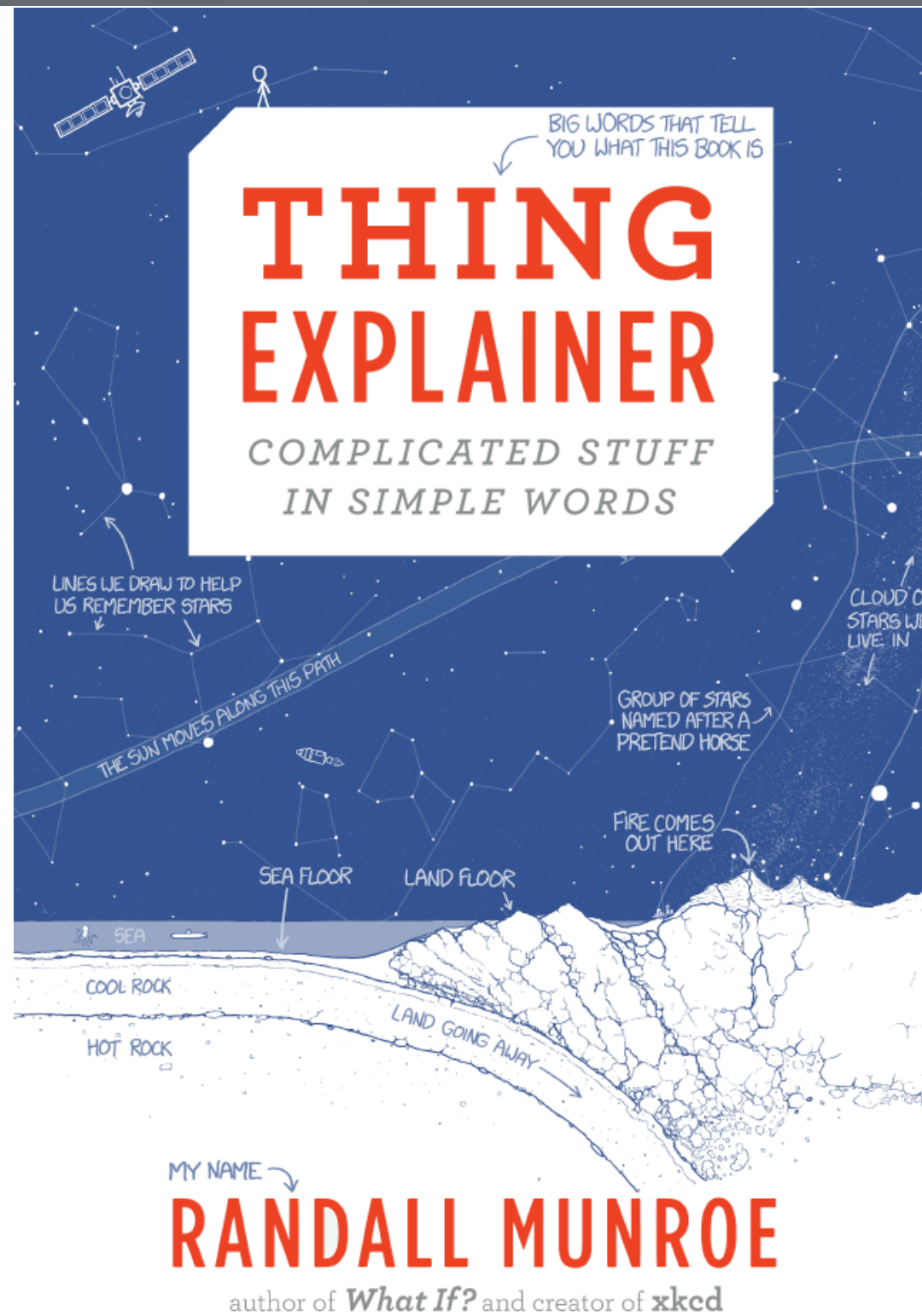
LECTURE 1

Introduction To Modelling & Simulation

- Understand difference between simulation and modelling
- Appreciate the reasons for doing each
- Understanding of course structure and assessment

MODELLING

Describing What X Is



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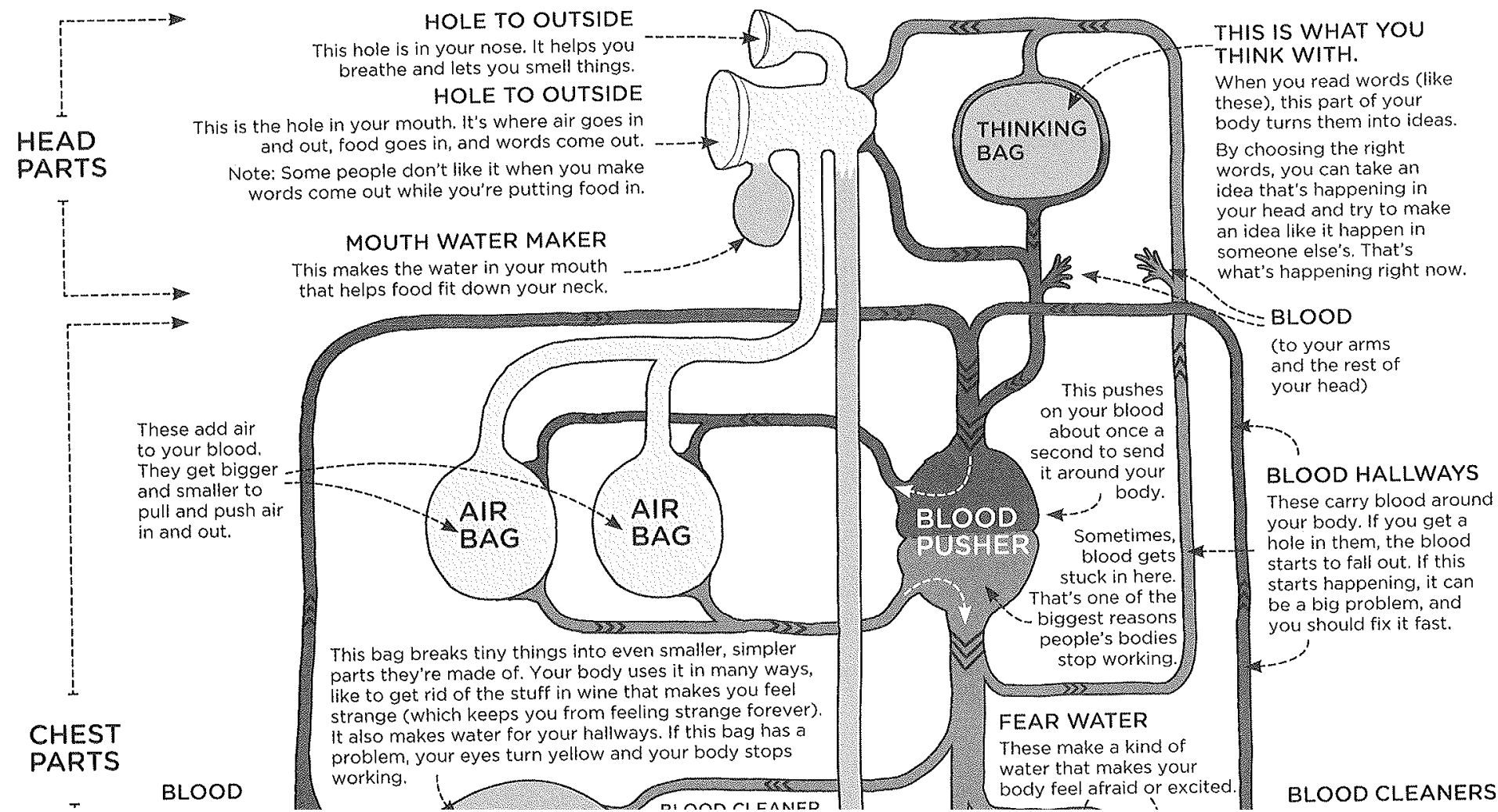
BAGS OF STUFF INSIDE YOU

This is a map of some of the different bags in your body and how they join together.

It doesn't show what they're really shaped like or how they're pushed together inside your body.

In that way, it's kind of like the colored maps in cities that tell you where trains go—it shows how the places are joined together, but not what they're shaped like or how far away from each other they are.

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Thing Explainer: Complicated Stuff in Simple Words, R. Munroe

SIMULATION

Predicting What Happens To X

SERIOUS SCIENTIFIC ANSWERS
to Absurd Hypothetical Questions



what if?



RANDALL MUNROE

creator of **xkcd**

SIMULATION

Predicting What Happens To X

HAIR DRYER

Q. What would happen if a hair dryer with continuous power were turned on and put in an airtight $1 \times 1 \times 1$ -meter box?

— Dry Paratroopa

A. A TYPICAL HAIR DRYER draws 1875 watts of power.

All 1875 watts have to go somewhere. No matter what happens inside the box, if it's using 1875 watts of power, eventually there will be 1875 watts of heat flowing out.

This is true of any device that uses power, which is a handy thing to know. For example, people worry about leaving disconnected chargers plugged into the wall for fear that they're draining power. Are they right? Heat flow analysis provides a simple rule of thumb: If an unused charger isn't warm to the touch, it's using less than a penny of electricity a day. For a small smartphone charger, if it's not warm to the touch, it's using less than a penny a *year*. This is true of almost any powered device.¹

But back to the box.

Heat will flow from the hair dryer out into the box. If we assume the dryer is indestructible, the interior of the box will keep getting hotter until the outer

¹ Though not necessarily those plugged into a second device. If a charger is connected to something, like a smartphone or laptop, power can be flowing from the wall through the charger into the device.

surface reaches about 60°C (140°F). At that temperature, the box will be losing heat to the outside as fast as the hair dryer is adding it inside, and the system will be in equilibrium.



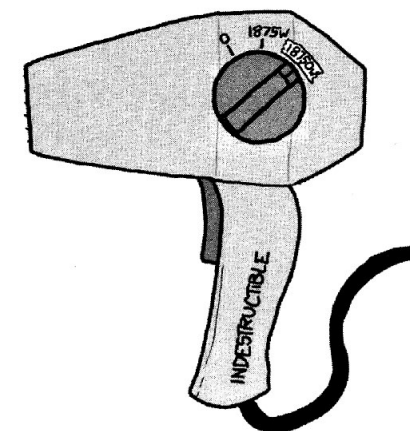
It's warmer than my parents! It's my new parents.

The equilibrium temperature will be a bit cooler if there's a breeze, or if the box is sitting on a wet or metallic surface that conducts away heat quickly.

If the box is made of metal, it will be hot enough to burn your hand if you touch it for more than five seconds. If it's wood, you can probably touch it for a while, but there's a danger that parts of the box in contact with the mouth of the hair dryer will catch fire.

The inside of the box will be like an oven. The temperature it reaches will depend on the thickness of the box wall; the thicker and more insulating the wall, the higher the temperature. It wouldn't take a very thick box to create temperatures high enough to burn out the hair dryer.

But let's assume it's an indestructible hair dryer. And if we have something as cool as an indestructible hair dryer, it seems like a shame to limit it to 1875 watts.



WHAT DOES INDUSTRY SAY?

The Importance Of Modelling & Simulation

How important is simulation and modelling to the industry for reducing costs, prototypes and testing?

“Very important. It is a big driver to reduce cost and improve product understanding. There is no chance to do this without simulation.”

Akin Keskin, Chief of Virtual Engine Design Systems, **Rolls-Royce**

“Simulation and Modelling is enormously important to the automotive industry today. This enormous engineering challenge can only be achieved by great engineers using the best simulation and modelling methods and tools to enable them to develop optimised and robust designs.”

Andy Richardson, Head of Simulation, Vehicle Engineering, **Jaguar Land Rover**

<http://events.imeche.org/speaker-interviews/60-seconds-with-akin-keskin-of-rolls-royce-and-andy-richardson-of-jaguar-land-rover>

WHY MODEL?

- Understanding
- Guide experiments
- Stimulate new questions
- Inform policy

TYPES OF MODEL

- Analytical vs numerical
- Mechanistic vs phenomenological
- Statistical vs deterministic

TYPES OF MODEL

- Analytical vs numerical
 - Closed form solutions (can be exact or approximate) vs
 - Solution approximated & solved using a method such as finite differences
- Mechanistic vs phenomenological
- Statistical vs deterministic

TYPES OF MODEL

- Analytical vs numerical
- Mechanistic vs phenomenological
 - Built from the “ground-up” using equations based on physical principles vs
 - Black box models such as machine learning techniques like neural networks
- Statistical vs deterministic

TYPES OF MODEL

- Analytical vs numerical
- Mechanistic vs phenomenological
- Statistical vs deterministic
 - One or more variables are probability density functions and associations between input and output are statistical vs
 - Direct repeatable link between an input and output

THE MAP IS NOT THE TERRITORY

Modelling Is Judicious Simplification

“... In that Empire, the Art of Cartography attained such Perfection that the map of a single Province occupied the entirety of a City, and the map of the Empire, the entirety of a Province. In time, those Unconscionable Maps no longer satisfied, and the Cartographers Guilds struck a Map of the Empire whose size was that of the Empire, and which coincided point for point with it. The following Generations, who were not so fond of the Study of Cartography as their Forebears had been, saw that that vast map was Useless, and not without some Pitilessness was it, that they delivered it up to the Inclemencies of Sun and Winters. In the Deserts of the West, still today, there are Tattered Ruins of that Map, inhabited by Animals and Beggars; in all the Land there is no other Relic of the Disciplines of Geography.”

purportedly from Suárez Miranda, Travels of Prudent Men, Book Four, Ch. XLV, Lérída, 1658
"On Exactitude in Science", J. L. Borges

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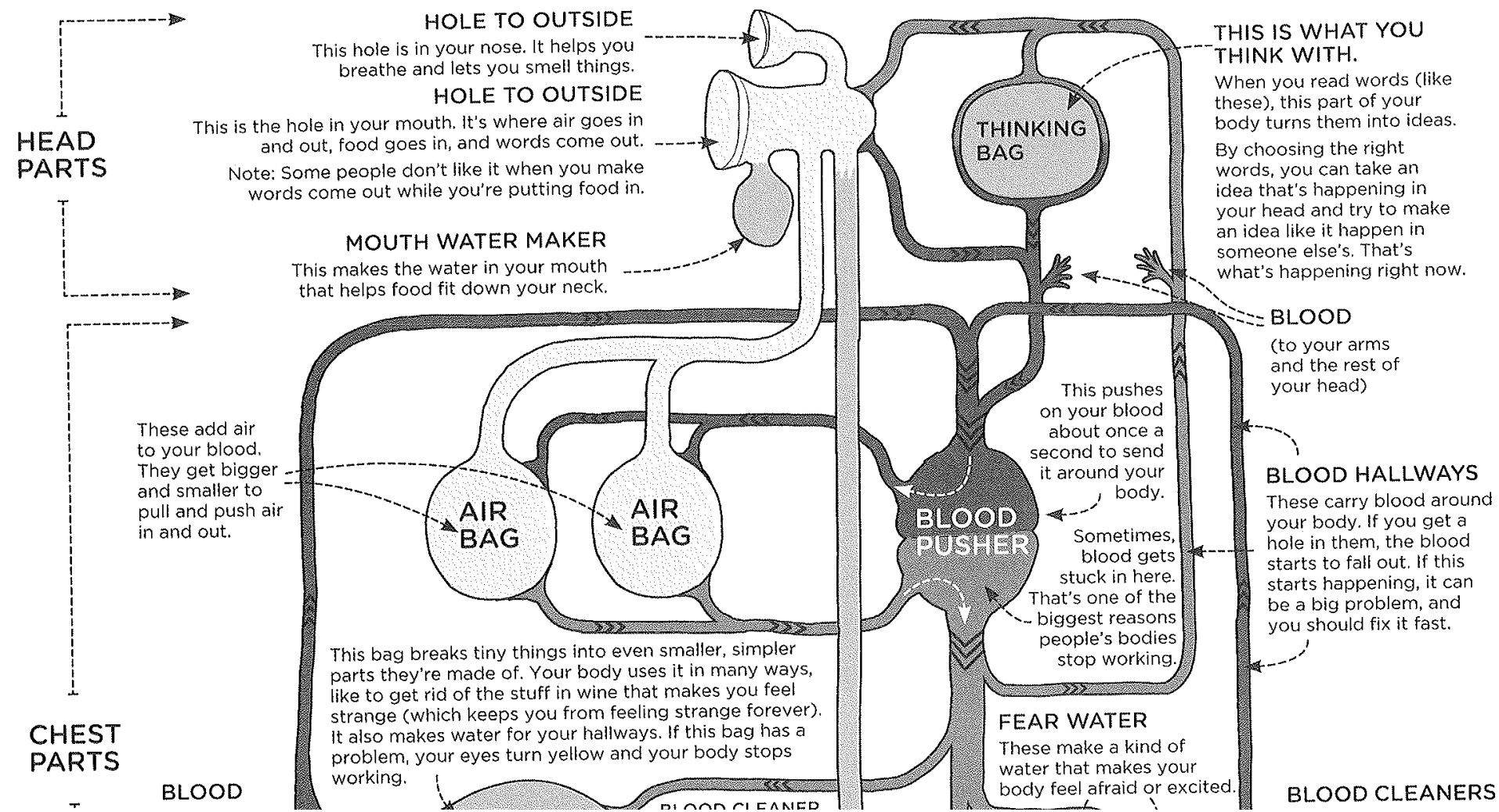
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WHY SIMULATE?

- Can reduce safety factor in designs
- Accelerates design optimisation
- Obtain quantities not accessible experimentally
- Extends the usefulness of experiments

EXAMPLES OF SIMULATION

- CFD analysis of aircraft performance
- Finite element analysis of structures
- Weather & climate forecasting

CHALLENGES OF SIMULATIONS

- Obtaining material parameters
- High computational run time
- Determining correct boundary conditions
- Access to suitable geometric model
- Availability of appropriate tool for physics of interest - might have to create your own
- Clear presentation of results - data overload!
- Appropriate numerical method - is it accurate?

THE MOTIVATION

In less than 12 months' time you will be graduate engineers...

2026

< Today >

January

M	T	W	T	F	S	S
29	30	31	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	1
2	3	4	5	6	7	8

February

M	T	W	T	F	S	S
26	27	28	29	30	31	1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	1
2	3	4	5	6	7	8

March

M	T	W	T	F	S	S
23	24	25	26	27	28	1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31	1	2	3	4	5

April

M	T	W	T	F	S	S
30	31	1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	1	2	3
4	5	6	7	8	9	10

May

M	T	W	T	F	S	S
27	28	29	30	1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31
1	2	3	4	5	6	7

June

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	1	2	3	4	5
6	7	8	9	10	11	12

July

M	T	W	T	F	S	S
29	30	1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31	1	2
3	4	5	6	7	8	9

August

M	T	W	T	F	S	S
27	28	29	30	31	1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31	1	2	3	4	5	6

September

M	T	W	T	F	S	S
31	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	1	2	3	4
5	6	7	8	9	10	11

October

M	T	W	T	F	S	S
28	29	30	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	1
2	3	4	5	6	7	8

November

M	T	W	T	F	S	S
26	27	28	29	30	31	1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	1	2	3	4	5	6

December

M	T	W	T	F	S	S
30	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31	1	2	3
4	5	6	7	8	9	10

COURSEWORK DELIVERABLES

Background

You are a recent graduate engineer and have just started working at a small engineering consultancy. The firm is trying to develop a reputation for technical excellence and is growing fast. As a result you must work with minimal supervision from your line manager.

In your coursework assignments you will be given different projects that will require computational models to be developed, in order to solve a technical problem for one of your firm's clients, often in a design context.

These models & software must be developed following good practices, be validated against standard test cases, and accompanied by appropriate documentary proof. Your reports will summarise the results and allow you to make justified recommendations to the client.

COURSEWORK DELIVERABLES

Outline

Part 1: Static MATLAB-based FEM modelling (Lectures 1 - 8)

Formative Assessment: 0% weighting BUT invaluable for feedback and completing CW1

Submission: Code and few pieces of evidence of code correctness e.g. test outputs. No report.

Part 2: Transient MATLAB-based FEM modelling (Lectures 9 - 14)

Coursework 1: 60% weighting

Submission: Code plus 3000 word report

Part 3: Block diagram & Simulink-based dynamic system modelling (Lectures 15 - 22)

Coursework 2: 40% weighting

Submission: Simulink model & code plus 2000 word report

ASSESSMENT TIMELINE

ME40064 DATES

Part 1: Static MATLAB-based FEM modelling (Lectures 1 - 8)

Formative Assessment: issued in Week 3

Due date: 4pm, Tuesday 4th November 2025

Part 2: Transient MATLAB-based FEM modelling (Lectures 9 - 14)

Coursework 1: issued on Tuesday 11th November 2025

Due date: 4pm, Tuesday 2nd December 2025

Part 3: Block diagram & Simulink-based dynamic system modelling (Lectures 15 - 22)

Coursework 2: issued on Monday 8th December 2025

Due date: 4pm, Friday 9th January 2026

SKILLS TO BE DEVELOPED

Coding

- implement complex algorithms in computer code using a robust and rigorous approach

Self-critical thinking

- toolkit for testing correctness of your own work

Communication

- convincing the reader of correctness of your recommendation through clear text and figures

ATTRIBUTES I'M LOOKING FOR

Rigour

- how you demonstrate correctness of code
- thorough exploration of results and parameter space
- quantification of modelling errors and numerical convergence

ATTRIBUTES I'M LOOKING FOR

Understanding

- physical interpretation of the results mean physically
- implications for design choices
- limitations of model and numerical approximation

ATTRIBUTES I'M LOOKING FOR

Communication

- clear, logical structure of report
- explanation of what is being done and why
- clear figures that support your argument

COURSE ADMINISTRATION

Live Online Sessions And In Person Time

Drop-In Sessions

- Online, every Thursday 11:00 -12:00.
- Link will be posted on Moodle page

In Person Time

Weekly one-hour IPT sessions - lecture. Mondays 10:15 - 11:05.

Weekly two-hour IPT sessions - mixture of lecture, discussion, and computer lab tutorial. Tuesdays 11:15 - 13:05.

Moodle Forum

Please post all technical queries to the Moodle forum. I will answer as many as I can but please help each other - it will benefit you all and likely be faster!

Any Other Business

For any other queries email me at a.n.cookson@bath.ac.uk . I will do my best to reply quickly BUT please keep in mind I am teaching ~80 students across ME40064 and ME52085.

FURTHER READING

- Why Model?, J. M. Epstein, SFI WORKING PAPER: 2008-09-040,
<http://jasss.soc.surrey.ac.uk/11/4/12.html>
- The Thing Explainer, R. Munroe
- What If?, R. Munroe